Avaljot Singh

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Research Interests

My research centers on a bidirectional interaction between **formal methods** and **artificial intelligence**. On one hand, I develop formal and declarative infrastructures for building trustworthy AI systems, with a focus on specification, soundness verification, and efficient execution of AI analyses and certification procedures. On the other hand, I study how learning and synthesis techniques can automate formal methods themselves, enabling automatic construction of verification tools, program analyses, and reasoning procedures. Together, these directions aim to improve the trustworthiness of AI systems and substantially reduce the human effort required to build and maintain formal verification frameworks.

Education

University of Illinois Urbana-Champaign — Urbana, IL
PhD in Computer Science (GPA: 4.0/4.0) — *Aug 2022 – December 2026 (Expected)*
Advisors: Prof. Gagandeep Singh, Prof. Charith Mendis

Indian Institute of Technology, Delhi — New Delhi, India
B.Tech + M.Tech in Computer Science (GPA: 9.5/10) — *July 2016 – May 2021*
Advisor: Prof. Sanjiva Prasad — Thesis: Algebraic techniques for network routing

Experience

Microsoft Research — Redmond, WA
Research Intern — Mentor: Shraddha Barke, Suman Nath — *May 2025 – Aug 2025*
- Studied failures in LLM-based AI agents by analyzing execution trajectories.
- Designed symbolic ways of agent execution to pinpoint failure modes and improve agent reliability.

Microsoft Research — Bangalore, India
Research Intern — Mentor: Rahul Sharma — *Mar 2021 – Jun 2021*
- Designed a domain-specific language and type system for Mixed-mode Multi-Party Computation applied to ML workloads.
- Formally proved correctness and cryptographic security guarantees for well-typed programs in the system.

Graviton Research Capital LLP — Gurugram, India
Quantitative Researcher — *Jun 2021 – Jul 2022*
- Developed and backtested systematic quantitative trading strategies using statistical modeling and data analysis.
- Built data pipelines and research infrastructure to evaluate signal quality across large financial datasets.

Uber — Hyderabad, India
Software Engineering Intern — *May 2020 – Jul 2020*
- Built an automated document transcription pipeline using text detection, recognition, and classification.
- Developed a novel text classification approach using graph isomorphisms to detect and encode textual structural features.

Cornell University — Ithaca, NY
Research Intern — Mentor: Prof. Nate Foster — *May 2019 – Jul 2019*
- Extended NetKAT, a formal framework for network programming, to support configurable edge devices.
- Defined syntax and semantics for Edge-NetKAT, pushing NetKAT program functionality to programmable network edges.

Hiroshima University — Hiroshima, Japan
Research Intern — Mentor: Prof. Idaku Ishii — *Jun 2018 – Jul 2018*
- Implemented a facial recognition system mounted on a 2-DOF mechanical tracking actuator for real-time security camera surveillance.
- Developed high-speed camera interfacing pipeline for real-time image synthesis and object tracking.

Research Projects

ConstraintFlow / ProveSound / Neuton — Formal Methods for AI

- Designed **ConstraintFlow**, a declarative DSL for specifying and verifying neural network certifiers, reducing implementation complexity significantly vs. hand-written implementations. *[SAS 2024]*
- Built **ProveSound**, an automated soundness verifier that formally checks certifier implementations against concrete neural network semantics. *[OOPSLA 2025]*
- Developed **Neuton**, a compiler and runtime translating ConstraintFlow specs into efficient executables for large-scale neural network verification. *[Under Submission]*

KV Cache Management for Agentic Workflows — Formal Methods for AI

- Developing principled methods for KV cache eviction and memory management tailored to the long-horizon, multi-step nature of LLM-based agentic workflows. *[Ongoing]*

RuleFlow AI for Compilers

- Built a system combining LLMs with compiler and program analysis techniques to automatically generate reusable Pandas and data-analysis optimizations. *[Under Submission]*

SAIL - Synthesizing Sound Abstract Interpreters AI for Formal Methods

- Developed a synthesis system that automatically generates provably-sound neural network certifiers using ConstraintFlow, its verifier, and LLM-guided search over certifier designs. *[PLDI 2026]*

Maestro - LLM-based Theorem Proving AI for Formal Methods

- Developed a unified operational formalism for LLM-based theorem-proving systems, supporting formal reasoning by guiding proof search and generating intermediate lemmas. *[VerifAI @ ICLR 2026]*

Publications

Under Review

- **Avaljot Singh***, Dushyant Bharadwaj*, Stefanos Baziotis, Kaushik Varadharajan, Charith Mendis. "RuleFlow: Generating Reusable Program Optimizations with LLMs." *Arxiv 2026.*
- Shraddha Barke, Arnav Goyal, Alind Khare, **Avaljot Singh**, Suman Nath, Chetan Bansal. "AgentRx: Diagnosing AI Agent Failures from Execution Trajectories." *Arxiv.*
- Isha Chaudhary, Vedaant Jain, **Avaljot Singh**, Kavya Sachdeva, Sayan Ranu, Gagandeep Singh. "Lumos: Let there be Language Model System Certification." *Arxiv.*
- **Avaljot Singh**, Yasmin Sarita, Aditya Mishra, Ishaan Goyal, Gagandeep Singh, Charith Mendis. "A Tensor-Based Compiler and a Runtime for Neuron-Level DNN Certifier Specifications." *Arxiv.*

Conference & Workshop Papers

- **Avaljot Singh***, Shaurya Gomber*, Yasmin Sarita, Jose Meseguer, Gagandeep Singh. "Unified Operational Formalism for LLM-based Theorem-proving Systems." *VerifAI @ ICLR 2026.*
- Yasmin Sarita, **Avaljot Singh**, Shaurya Gomber, Gagandeep Singh, Mahesh Viswanathan. "Efficient Ranking Function-Based Termination Analysis with Bi-Directional Feedback." *ESOP 2026.*
- Qiuhan Gu, **Avaljot Singh**, Gagandeep Singh. "Cost-Driven Synthesis of Sound Abstract Interpreters." *PLDI 2026.*
- **Avaljot Singh**, Yasmin Sarita, Charith Mendis, Gagandeep Singh. "Automated Verification of Soundness of DNN Certifiers." *OOPSLA 2025.*
- **Avaljot Singh**, Yasmin Sarita, Charith Mendis, Gagandeep Singh. "ConstraintFlow: A DSL for Specification and Verification of Neural Network Analyses." *SAS 2024.*
- Debangshu Banerjee, **Avaljot Singh**, Gagandeep Singh. "Interpreting Robustness Proofs of Deep Neural Networks." *ICLR 2024.*
- Ashitabh Misra, Madhav Agrawal, Tomoyoshi Kimura, Jinyang Li, **Avaljot Singh**, Arham Jain, Tarek Abdelzaher. "Adaptive Quantization of CNNs for Spectrogram-Based Foreground Activity Classification." *IJCNN 2026.*

Journal

- Gagandeep Singh, Jacob Laurel, Sasa Misailovic, Debangshu Banerjee, **Avaljot Singh**, Changming Xu, Shubham Ugare, Huan Zhang. "Safety and Trust in Artificial Intelligence with Abstract Interpretation." *Foundations and Trends in Programming Languages, 2025.*

Awards & Achievements

Bronze Medal — SRC @ PLDI 2024 (ConstraintFlow)

Outstanding Paper Award — WFVML @ ICML 2023 (Interpreting Robustness Proofs)

All India Rank 141 — IIT Joint Entrance Examination (JEE Advanced), 2016

IIT Delhi Semester Merit Award — Top 7% in department, 7 consecutive semesters

KVPY Fellow — Selected by IISc Bangalore, 2015; **NTSE Scholar** — Selected by CBSE Delhi, 2013

Teaching & Service

Teaching Assistant — CS477: Formal Software Development Methods, UIUC (Spring 2024)

Teaching Assistant — Analysis and Design of Algorithms; Intro to Functional Programming; Programming Languages; Intro to CS, IIT Delhi (2019–2021)

Reviewer — Formal Methods in System Design 2024, VerifAI@ICLR 2026

Artifact Evaluation — ECOOP 2025, CAV 2025, PLDI 2026